

Appointment Scheduling Engine

Duration: 6 Hours

Technology:

Laravel 12

PHP 8.2+

MySQL

React (Frontend UI)

Scenario

Develop an Appointment Scheduling Module for a hospital application.

Build a user-friendly admin interface to manage Departments, Doctors, Patients, and Appointments.

The system should efficiently schedule appointments while preventing scheduling conflicts. The application should also be designed to perform well with large datasets.

Requirements

1. Database

Design and create the required database structure.

Your database design should be properly normalized and include appropriate relationships.

2. Sample Data

Create sample data for:

Departments

Doctors

Patients

Use Laravel Factories and Seeders to generate the data.

3. Appointment Management

The administrator should be able to:

Create an appointment.

Select a Department.

Select a Doctor.

Select a Patient.

Choose Appointment Date & Time.

Specify Appointment Duration (in minutes).

Set Appointment Status:

Scheduled

Completed

Cancelled

Edit existing appointments.

Cancel appointments.

4. Business Rules

Implement the following business rules:

A doctor cannot have overlapping appointments.

A patient cannot have overlapping appointments.

Appointments cannot be scheduled in the past.

Cancelled appointments should not block future bookings.

Appointment duration must be greater than zero.

Display available appointment slots for the selected doctor on a given date.

5. Appointment Listing

Display a paginated list of appointments including:

Appointment ID

Patient Name

Doctor Name
Department
Appointment Date & Time
Duration
Status
Provide filters for:
Doctor
Department
Appointment Date
Status
Sorting by appointment date is recommended.

6. API

Create an API endpoint to retrieve available appointment slots for a selected doctor on a given date.

Example:

GET /api/doctors/{doctor}/available-slots?date=2026-06-20

The API should return:

Available appointment slots
Appropriate HTTP status codes
Consistent JSON response format
Proper validation messages
Resource Not Found response when applicable

7. Large Data Handling

The system should be designed to work efficiently with a large volume of data.

Generate sample data using Laravel Factories and Seeders.

Minimum dataset:

20 Departments

500 Doctors

100,000 Patients

1,000,000 Appointments

The candidate does not need to manually create these records. Use Factories and Seeders to generate the data.

Performance Expectations

The application should be optimized for handling large datasets.

8. Frontend

Build the admin interface using React. The UI should be responsive, user-friendly, and provide a good user experience.

Deliverables

Submit the following:

Laravel project source code

Database migrations

Model Factories

Seeders

API endpoint